Rahul Maurya

Software Engineer · Backend & Distributed Systems

Melbourne, Australia · +61 401 009 310 · maurya2811@gmail.com · Portfolio · LinkedIn · GitHub · LeetCode

Work rights: Subclass 482 (Skills in Demand) visa holder; requires employer nomination transfer only

SUMMARY

Backend software engineer with 2.5+ years building and running event-driven Java / Spring Boot microservices for warehouse automation (WES / WCS / ASRS) that enterprise clients operate 24/7. Currently forward-deployed on a live distribution centre in Melbourne, where I raised goods-to-person picking throughput by 50% and closed the gap to a contracted SLA. Strong in distributed systems, messaging (RabbitMQ, Kafka), Kubernetes and production observability, with hands-on LLM tooling experience.

EXPERIENCE

Software Engineer · Addverb Australia

Melbourne, Australia — Jan 2026 – Present

- Forward-deployed on a **20+ microservice** warehouse execution platform (Java 17, Spring Boot, RabbitMQ, MongoDB, SQL Server, Kubernetes / Helm) for a tier-1 global retail brand's distribution centre; own change requests from design through production rollout.
- Raised goods-to-person picking throughput from **800 to 1,200 totes/hr (+50%)** by tracing per-stage dwell time and fixing work-release and allocation logic, closing the gap to the contracted SLA and unblocking a **seven-figure (AUD)** contractual milestone.
- Built a cross-service irregular-event store (RabbitMQ to MongoDB, correlation-id indexed) with Grafana views, replacing manual log searches across pods with a single correlated query for root-cause analysis.
- Segregated competing business flows on shared automation, improving client operational efficiency by **30%**.

Software Development Engineer · Addverb Technologies

Noida, India — Apr 2024 – Dec 2025

- Decomposed monolithic modules of a Warehouse Control System into **12+ Spring Boot microservices** over RabbitMQ, cutting p95 API latency from **800 ms to 600 ms (−25%)**.
- Built a real-time event pipeline on MongoDB with tuned MySQL / SQL Server queries, processing **10,000+ events/min with zero data loss** (idempotent consumers, dead-letter queues, replay).
- Sustained **99.6% uptime** across production sites through Docker / Kubernetes deployments and Jenkins CI/CD.
- Built an LLM-powered inventory assistant (RAG plus read-only tool calls) that auto-resolves **75%+** of operator queries, reducing support tickets by **40%**.
- Wrote a Python tool that migrates Grafana dashboards across 3 network environments (200+ dashboards), cutting each migration from **8 hours to 20 minutes**.
- Delivered monitoring and hardware health dashboards that cut incident response time by **60%**; shipped WMS features to **5+ enterprise clients** in Scrum teams with QA, DevOps and Product.

Software Engineering Intern · NEC Corporation India

Noida, India — Oct 2023 – Jan 2024

- Built Java / Spring Boot microservice components and integrated messaging for inter-service communication.

SELECTED PROJECTS

Robotic Shuttle Integration & Telemetry · US healthcare logistics provider

Sep 2025 – Dec 2025

- Integrated third-party robotic shuttles and lifts through TCP / REST adapters; MongoDB telemetry pipeline at **99.6% capture** across 50+ devices; health dashboards cut downtime diagnosis by **70%**.

Tote Routing Optimisation · D2C retailer fulfilment centre

Jun 2025 – Aug 2025

- Graph-based routing in Java (Dijkstra plus Yen's K-shortest alternates on congestion-weighted edges): **−200 ms** per routing decision, **−15%** path resolution time, **+12%** pilot throughput.

Deep-Lane ASRS (5-deep, 9-level) · Tyre manufacturing plant

Oct 2024 – May 2025

- Orchestrated cranes, shuttles, duplex lifts and conveyors for **500+ daily transactions**; cut client interface testing from **3 months to 3 weeks** by helping QA and Process teams complete interface development.

Manual-Format WMS · Large Indian grocery retailer

2024

- Fixed **70+ critical UAT defects** and ran 14 demos for senior client leadership; received a formal note of appreciation from the project manager.

RFP Management System · B.Tech capstone

Jul 2023 – Feb 2024

- Java / Spring Boot procurement platform with role-based access for 3+ personas and 15+ configurable report templates.

TECHNICAL SKILLS

Languages: Java 17, SQL, Python

Backend: Spring Boot, Spring Data JPA, REST APIs, microservices, event-driven architecture, multithreading & concurrency

Messaging & Data: RabbitMQ, Apache Kafka, Redis, MongoDB, MySQL, SQL Server

Cloud & DevOps: Docker, Kubernetes, Helm, Jenkins, Git, CI/CD; AWS (EC2, EKS, S3, RDS, IAM) working knowledge

Observability: Grafana, Jaeger, distributed logging, event-store based root-cause analysis

AI: LLM APIs, retrieval-augmented generation (RAG), tool-calling agents, prompt engineering

Familiar with: gRPC, GraphQL, OAuth2 / JWT, Terraform, LangChain

EDUCATION

B.Tech, Computer Science & Engineering

Raj Kumar Goel Institute of Technology, Ghaziabad (AKTU) · CGPA 8.89 / 10 — 2020 – 2024

CERTIFICATIONS & LEARNING

- AWS Certified Solutions Architect – Associate (in preparation)
- Generative AI with Large Language Models – DeepLearning.AI & AWS (Coursera)
- Software Design and Architecture Specialization – University of Alberta (Coursera)